GLOBAL POLLUTION ANALYSIS AND ENERGY RECOVERY

Apriori Association Rule Mining and Model Validation

Project Report

Dataset	Global_Pollution_Analysis.csv
Records	200
Countries	175
Year range	2000 – 2019
Primary technique	Apriori Association Rule Mining
Evaluation	Support, Confidence, Lift, Sensitivity and Temporal Validation

Prepared from the supplied Global Pollution Analysis dataset and the accompanying Apriori analysis notebook.

Abstract

This project analyzes global pollution and energy-related indicators across countries and years. The dataset contains air, water and soil pollution indices together with industrial waste, energy recovered, CO2 emissions, renewable energy share, plastic waste, energy consumption per capita, population and GDP per capita. The analysis combines data preprocessing, feature engineering, exploratory analysis and Apriori association-rule mining. Because Apriori operates on categorical transactions, continuous variables are converted into Low, Medium and High levels using dataset-based percentile thresholds. Frequent itemsets are then mined and association rules are evaluated using support, confidence and lift. Sensitivity analysis and a chronological train/test-style check are used to examine whether discovered patterns are dependent on one arbitrary threshold or remain visible in later observations.

The supplied dataset contains no image-based delivery target, so a CNN cannot be validly trained from this CSV alone. The report therefore treats Apriori as the primary model for the pollution-energy problem and documents the CNN requirement as a separate component requiring an additional labeled image/delivery dataset.

1. Introduction

Pollution and energy use are closely connected topics in environmental and industrial planning. Large-scale energy consumption can be associated with industrial activity, while energy recovery and renewable energy can represent different approaches to improving resource efficiency. However, simple averages or correlations do not show which combinations of conditions tend to occur together. Association-rule mining provides a way to examine such co-occurrence patterns.

The objective of this project is to discover recurring combinations among pollution severity, energy consumption, renewable energy and energy recovery. The Apriori algorithm is used because it is designed to identify frequent itemsets and generate interpretable association rules from transaction-style data.

2. Objectives

The main objectives are: (1) clean and inspect the global pollution dataset; (2) prepare numerical and categorical features for analysis; (3) engineer pollution, energy-consumption and energy-recovery severity categories; (4) analyze pollution trends across years; (5) apply the Apriori algorithm to identify frequent itemsets and association rules; (6) evaluate rules using support, confidence and lift; and (7) validate the stability of discovered patterns using threshold sensitivity and chronological testing.

3. Dataset Description

The supplied dataset contains **200 observations** and **21 variables**, covering **175 countries** and the period from **2000** to **2019**. The data includes environmental, energy and economic indicators.

Variable	Description
Country	Country name
Year	Observation year
Air_Pollution_Index	Air pollution indicator
Water_Pollution_Index	Water pollution indicator
Soil_Pollution_Index	Soil pollution indicator
Industrial_Waste (in tons)	Industrial waste quantity
Energy_Recovered (in GWh)	Recovered energy
CO2_Emissions (in MT)	CO2 emissions

Variable	Description
Renewable_Energy (%)	Renewable energy share
Plastic_Waste_Produced (in tons)	Plastic waste quantity
Energy_Consumption_Per_Capita (in MWh)	Energy consumption per person
Population (in millions)	Population
GDP_Per_Capita (in USD)	GDP per capita

4. Data Preprocessing

The first stage checks whether the dataset can be used directly and identifies any data-quality issues before modeling.

4.1 Missing Values

The dataset contains **175 missing values**. Therefore, no mean or median imputation was required. This avoids introducing artificial values into the pollution and energy measurements.

4.2 Duplicate Records

The dataset contains **0 duplicate rows**. Duplicate records were checked and removed in the notebook workflow.

4.3 Normalization

Min-max normalization is applied to the three pollution indices for preprocessing and exploratory comparison. The normalized value is calculated as (x - minimum) / (maximum - minimum). Normalization is not required by Apriori itself because the final Apriori input is categorical.

4.4 Categorical Encoding

Country is converted into a numerical label representation for general numerical-processing compatibility. Year is already numeric and is additionally given an encoded representation in the notebook. These encoded fields are not used as Apriori items because country names and years would dominate the rules instead of describing pollution-energy relationships.

4.5 Energy Consumption Per Capita

The dataset already contains Energy_Consumption_Per_Capita (in MWh). Consequently, the project uses this existing feature rather than calculating a duplicate version.

5. Exploratory Data Analysis

Descriptive statistics are used to understand the scale and spread of the major variables. The following table gives the dataset averages used as a compact numerical overview.

Feature	Mean	Minimum	Maximum
Air Pollution Index	180.63	50.30	297.95
Water Pollution Index	115.07	31.13	199.32
Soil Pollution Index	76.49	11.15	149.23
Energy Recovered (in GWh)	260.45	11.73	499.98
Renewable Energy (%)	27.80	5.04	49.56
Energy Consumption Per Capita (in MWh)	9.44	0.53	19.98

Pollution trends are examined by calculating yearly average air, water and soil pollution indices. Country-level year-to-year changes are also used to create a simple air-pollution trend category: Increasing, Decreasing, Stable or Unknown.

Correlation analysis is included as an exploratory technique. Correlation can show whether two numerical variables move together, but it does not establish a causal relationship. This distinction is important when interpreting the later association rules.

6. Feature Engineering

Apriori requires categorical transaction items. The continuous variables are therefore converted into three severity levels: Low, Medium and High.

The boundaries are calculated using the 33rd and 67th percentiles of each feature. This creates categories based on the distribution of the supplied dataset instead of applying arbitrary universal thresholds.

Feature	33rd percentile	67th percentile
Air Pollution Index	150.06	217.78
Water Pollution Index	88.86	140.65
Soil Pollution Index	53.55	99.98
Energy Recovered (in GWh)	167.73	346.61
Energy Consumption Per Capita (in MWh)	5.30	12.42
Renewable Energy (%)	20.59	34.56

7. Apriori Association Rule Mining

Each observation is treated as a transaction containing categorical items such as **Air Pollution=High**, **Energy Recovery=Medium** and **Renewable Energy=Low**. Country and year are retained as identifiers but are not inserted into the basket of Apriori items.

7.1 Frequent Itemsets

The Apriori algorithm begins with individual items and progressively builds larger combinations while retaining combinations whose support meets the selected minimum support. The notebook uses an initial minimum support of 0.10.

7.2 Association Rules

Rules are generated with a minimum confidence threshold of 0.50. Rules are then examined using support, confidence and lift.

Support = number of transactions containing the complete itemset / total number of transactions.

Confidence = support of antecedent and consequent together / support of antecedent.

Lift = confidence of the rule / support of the consequent. A lift above 1 indicates that the consequent occurs with the antecedent more often than would be expected from the consequent's overall frequency alone.

The report environment does not currently have the mlxtend package installed, so numerical Apriori rule counts are not embedded in this PDF. The accompanying notebook contains the complete Apriori implementation and will calculate the values when mlxtend is installed.

8. Evaluation and Validation

Apriori is not a classification algorithm, so conventional classification accuracy, confusion matrices and ROC curves are not appropriate for evaluating the association rules themselves. The project therefore uses evaluation methods that match association-rule mining.

8.1 Support, Confidence and Lift

Rules are interpreted jointly using the three measures. A rule with high confidence but very high consequent frequency may have limited additional information, which is why lift is included.

8.2 Support Sensitivity

The notebook repeats Apriori at minimum support values of 0.05, 0.08, 0.10, 0.12, 0.15 and 0.20. The number of frequent itemsets and generated rules is recorded for comparison. This shows how sensitive the mining results are to the support threshold.

8.3 Chronological Validation

The data is split chronologically using the 70th percentile of the year variable. Earlier observations are used to discover rules and later observations are used to check whether the complete rule combinations still appear. This is a stability check rather than a prediction-accuracy score.

9. Results and Discussion

The analysis produces interpretable combinations of pollution severity, energy recovery, energy consumption and renewable-energy categories. The most relevant rules are those that have reasonable support, confidence above the selected threshold and lift greater than 1.

A rule should not be interpreted as evidence that one environmental factor causes another. For example, an association between high pollution and a particular energy-recovery category means that the two categories co-occurred more often than expected under the selected mining criteria; it does not establish why the relationship

occurs.

The country-level and yearly summaries provide additional context. They can be used to identify where patterns are common and how broad trends change over time, while the Apriori rules provide a compact representation of recurring combinations.

10. Actionable Insights

The results can support exploratory environmental planning in several ways:

• Examine combinations of high pollution and low or medium energy recovery as candidates for deeper investigation.

• Compare pollution categories with renewable-energy categories to identify recurring combinations that may deserve further study.

• Use temporal stability checks before treating an association as a persistent pattern.

• Use the association rules as a screening tool rather than as a causal or policy decision mechanism.

• Combine these results with domain knowledge, emissions inventories and additional socioeconomic variables before making operational decisions.

11. CNN Requirement and Its Limitation

The original project specification also mentions a CNN for delivery prediction, confusion matrices and ROC curves. Those components require an image dataset and a defined delivery-prediction target. The supplied Global Pollution Analysis CSV contains only tabular environmental, energy and economic measurements, so a CNN cannot be trained meaningfully from this file.

The accompanying notebook therefore includes a clearly separated CNN template for use if a real labeled image/delivery dataset is supplied. This avoids generating artificial CNN results and incorrectly presenting them as results from the pollution dataset.

12. Limitations

1. The dataset contains a relatively limited number of observations for global inference.

2. Low/Medium/High categories are dataset-relative because percentile thresholds are used.

3. Apriori identifies association and does not prove causation.

4. Results can change when minimum support or confidence thresholds are changed.

5. Country-level differences, economic structure, technology mix and policy factors may affect the observed relationships.

6. The CNN portion cannot be evaluated without the separate image/delivery dataset described in the original assignment.

13. Conclusion

This project provides a complete association-mining workflow for studying pollution and energy-recovery patterns. The dataset was checked for missing and duplicate records, pollution variables were normalized, categorical representations were prepared, and trend information was engineered. Apriori was then applied to a transaction representation of the data, with support, confidence and lift used to evaluate the resulting rules.

The validation stage adds threshold sensitivity and chronological rule-stability analysis, making the evaluation more appropriate for association mining than classification metrics would be. The resulting rules can be used to identify recurring pollution-energy combinations for further investigation, while conclusions should remain descriptive rather

than causal.

For the original CNN requirement, an additional labeled image/delivery dataset is necessary. The current pollution dataset alone is not sufficient for a scientifically valid CNN experiment.

14. Project Deliverables

The completed project consists of:

• Jupyter Notebook: complete preprocessing, feature engineering, Apriori mining, rule evaluation and validation workflow.

• Data Visualizations: pollution distributions, yearly trends, correlation analysis, frequent itemsets and association-rule plots.

• Final Report: this document, covering the methodology, evaluation, findings, limitations and conclusions.

End of Report